Objective Type

1. The measure attributes store quantitative information, which can be aggregated upon.
2. The value zero for a similarity measure indicates no similarity between two objects.
3. The data cube shows how different attributes of data are arranged in the data model.
4. Bitmap indexes are not highly efficient for quickly updating large datasets.
5. Precision, measures what percentage of the retrieved documents are actually relevant to the query.
6. In a source-driven architecture for gathering data, the data sources transmit new information, either continually (as transaction processing takes place), or periodically (nightly, for example).
7. The relations in a data warehouse schema can usually be classified as fact tables and dimension tables.
8. SPARQL is a query language designed to query RDF data.
9. Fact tables record information about individual events, such as sales, and are usually very large.
10. Chi-Square test helps us determine if there is a significant relationship between two categorical variables and the target variable.
11. The lines outside the box, extending to the highest and lowest values in a box plot are called as whiskers.
12. Cosine similarity is commonly used in text analysis to measure the similarity between documents based on the frequency of words or phrases they contain.
13. Sparse columns reduce the space requirements for null values at the cost of more overhead to retrieve non-NULL values.
14. Resource Description Framework (RDF) is a standard model for exchanging data on the web by describing resources using subject-predicate-object "triples".
15. MOLAP uses multidimensional array structures to store data for online analytical processing.
16. OLAP server is responsible for querying and analyzing data stored in the data warehouse, providing users with multidimensional views of the data for interactive analysis and reporting.
17. Correlation analysis is a statical method that is used to determine the strength and direction of the relationship between two continuous variables.
18. Correlation Analysis is used to study quantitative variables.
19. A binary attribute is asymmetric if the outcomes of the states are not equally important.
20. Fact table is the central table in a data warehouse's dimensional model, storing quantitative data from business processes.
21. The cube has 3 dimensions and each dimension has 4 levels (including all), the total number of cuboids will this cube contain 125 $cuboid$s.

22. Independent data marts are sourced from data captured from one or more operational systems or external information providers, or from data generated locally within a particular department or geographic area.
23. Dependent data marts are sourced directly from enterprise data warehouses.
24. ROLAP uses relational tables to store data for online analytical processing.
25. In a destination-driven architecture, the data warehouse periodically send requests to the data sources to retrieve new information.

MCQ

1. If the mean of a dataset is greater than the median, what is the shape of the distribution likely to be?
   A. Skewed right
   B. Skewed left
   C. Symmetric
   D. Bimodal
2. The binary attributes with both the categories having equal importance are called as _____.
   a) Symmetric
   b) Asymmetric
   c) Ordinal
   d) Numeric
3. Which of the following cannot be appropriately represented as a continuous attribute?
   a) Height
   b) Weight
   c) Time
   d) ZIP codes
4. Which of the following is not the preferred attribute type for quantile plots?
   a) Nominal
   b) Ordinal
   c) Interval-scaled
   d) Ratio-scaled
5. Which of the following is not true about scatter plots?
   a) It is used to identify relationship between attributes
   b) It is used in the case of univariate distribution
   c) It is used to identify outliers
   d) It is used to identify clusters
6. The median is also known as _____
   a) 2-quantile
   b) 3-quantile

c) 4-quantile
d) 5-quantile

7. Which of the following is true about interquartile range?
   a) It is the distance between first and third quartiles
   b) It is the distance between first and fourth quartiles
   c) It does not measure the spread of the data
   d) It is very sensitive to outliers
8. Which of the following are the components of five-number summary in the case of skewed data?
   a) Minimum, Q1, Median, Q3, Maximum
   b) Minimum, Q1, Mode, Q3, Maximum
   c) Minimum, Q1, Mode, Q3, Maximum
   d) Minimum, Q1, Mode, Q2, Maximum
9. The lines outside the box, extending to the highest and lowest values in a box plot are called as _____
   a) Medians
   b) Quartiles
   c) Quintiles
   d) Whiskers
10. Which of the following does not describe the properties of a data object?
    a) Attributes
    b) Features
    c) Dimensions
    d) Instances
11. Which of the following refers to the distribution of data based on two variables?
    a) Bivariate distribution
    b) Univariate distribution
    c) Time series
    d) Association rules
12. Which of the following cannot be the value of a nominal attribute?
    a) Names of things
    b) Symbols
    c) Numbers intended for quantitative use
    d) Numbers not intended for quantitative use
13. Which of the following is not a proximity measure?
    a) Similarity measures
    b) Dissimilarity measures
    c) Distance measures
    d) Probability measures

14. Which of the following refers to the dissimilarity matrix of objects?
    a) Attribute by object structure
    b) Object by object structure
    c) Clustered group structure
    d) Attribute by attribute structure
15. Which data warehouse schema arranges data in a star-like structure with a central fact table surrounded by dimension tables?
    a) Snowflake Schema
    b) Star Schema
    c) Constellation Schema
    d) Hybrid Schema
16. Which term refers to a subset of a data warehouse that focuses on specific subject areas for a particular group of users?
    a) Data Warehouse
    b) Data Mart
    c) Data Cube
    d) Data Schema
17. Which data warehousing component is responsible for storing information about the structure and contents of the data warehouse?
    a) Data Warehouse
    b) OLAP Server
    c) Metadata Repository
    d) Data Mart
18. What is the first step in the data analytics process?
    a) Data modeling
    b) Data collection
    c) Data visualization
    d) Decision making
19. If for an object, n be the number of attributes, m be the number of matches in case the state of two objects is same, then which of the following is true regarding the similarity of nominal attributes?
    a) Similarity = n/m
    b) Similarity = m/n
    c) Similarity = 2 * m * n
    d) Similarity = m * n
20. Task of inferring a model from labeled training data is called
    a) Unsupervised learning
    b) Supervised learning
    c) Reinforcement learning

Define following terms.

1. Data Warehouse
   A data warehouse is a repository (archive) of information gathered from multiple sources, stored under a unified schema, at a single site.

2. Data Quality
   Data have quality if satisfy the requirements of the intended use. Data quality assess accuracy, completeness, consistency, timeliness, believability, and interpretability.

3. Midrange
   A midrange can also be used to assess the central tendency of a numeric data set. It is the average of the largest and smallest values in the set.
4. Scatter plot
   A scatter plot is one of the most effective graphical methods for determining if there appears to be a relationship, pattern, or trend between two numeric attributes.

5. Data Transformation
   A function that maps the entire set of values of a given attribute to a new set of replacement values that is each old value can be identified with one of the new values.

6. Data Cube
   A data warehouse is based on a multidimensional data model which views data in the form of a data cube.

7. Resource Description Framework
   The Resource Description Framework (RDF) is a data representation standard based on the entity-relationship model. RDF consists of "triples," or statements, with a subject, predicate, and object.

8. What is data cube measure?
   A data cube measure is a numeric function that can be evaluated at each point in the data cube space. A measure value is computed for a given point by aggregating the data corresponding to the respective dimension–value pairs defining the given point.

9. What is data mining?
   Data mining is the process of discovering interesting patterns and knowledge from large amounts of data. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.

10. What is Resource Description Framework (RDF) and how to represent data in RDF model?

TheResource Description Framework (RDF) is a data representation standard based on the entity-relationship model. The RDF model represents data by a set of triples that are in one of these two forms:
1. (ID, attribute-name, value)
2. (ID1, relationship-name, ID2)
The first attribute of a triple is called its subject, the second attribute is called its predicate, and the last attribute is called its object.

11. What is the most popular data model for a data warehouse and which form exist?

    The most popular data model for a data warehouse is a multidimensional model, which can exist in the form of a star schema, a snow flake schema, or a fact constellation schema.

12. What is Attribute subset selection and goal of it?

    Attribute subset selection reduces the data set size by removing irrelevant or redundant attributes (or dimensions). The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.
13. How to measure retrieval effectiveness in an information-retrieval system able to answer queries?
    Two metrics are used to measure how well an information-retrieval system is able to answer queries. The first, precision, measures what percentage of the retrieved documents are actually relevant to the query. The second, recall, measures what percentage of the documents relevant to the query were retrieved.

14. What is the difference between data warehouse and data mart?

    A data warehouse collects information about subjects that span the entire organization such as customers, items, sales, assets, and personnel, and thus its scope is enterprise-wide. A data mart, on the other hand, is a department subset of the data warehouse that focuses on selected subjects, and thus its scope is department wide.

15. List the major steps involved in data preprocessing.

    The major steps involved in data preprocessing, namely, data cleaning, data integration, data reduction, and data transformation.

Write short notes.

1. What is Data Mart from the architecture point of view?

   Data mart contains a subset of corporate-wide data that is of value to a specific groups of users. The scope is confined to specific selected groups, such as marketing data mart. It may confine its subjects to customer, item and sales. The data contained in data marts tend to be summarized. Independent vs. dependent (directly from warehouse) data mart.

2. What is meta data and what are contained in its?

   Meta data is the data defining warehouse objects. It stores:

- **Description of the structure of the data warehouse** such as schema, view, dimensions, hierarchies, derived data defn, data mart locations and contents
- **Operational meta-data** such as data lineage, currency of data , monitoring information
- **The algorithms** used for summarization
- **The mapping from operational environment** to the data warehouse
- **Data related to system performance** such as warehouse schema, view and derived data definitions
- **Business data** such as business terms and definitions, ownership of data, charging policies

3. Comparison of OLTP and OLAP (any Five)

Feature	*OLTP*	*OLAP*
Characteristic	operational processing	informational processing
Orientation	transaction	analysis
User	clerk, DBA, database professional	knowledge worker (e.g., manager, executive, analyst)
Function	day-to-day operations	long-term informational requirements decision support
DB design	ER-based, application-oriented	star/snowflake, subject-oriented
Data	current, guaranteed up-to-date	historic, accuracy maintained over time
Summarization	primitive, highly detailed	summarized, consolidated
View	detailed, flat relational	summarized, multidimensional
Unit of work	short, simple transaction	complex query
Access	read/write	mostly read
Focus	data in	information out
Operations	index/hash on primary key	lots of scans
Number of records accessed	tens	millions
Number of users	thousands	hundreds
DB size	GB to high-order GB	$\geq$ TB
Priority	high performance, high availability	high flexibility, end-user autonomy
Metric	transaction throughput	query throughput, response time

4. Why data integration is needed in data preprocessing?
   Data integration combines data from multiple sources to form a coherent data store. Careful integration can help reduce and avoid redundancies and inconsistencies in the resulting data set. This can help improve the accuracy and speed of the subsequent data mining process. The resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplication detection, and data conflict detection contribute to smooth data integration.

5. Why data reduction is needed and what are the reduction strategies?
   Data Reduction obtain a reduced representation_of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results. A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set. Reduction strategies are
   -Data reduction
   -Numerosity reduction
   -Data compression

6. Differences between wide column and sparse column.
   Wide column representation: allow each tuple to have a different set of attributes, can add new attributes at any time.
   Sparse column representation: schema has a fixed but large set of attributes, by each tuple may store only a subset.

7. Define four differences views regarding a data warehouse design.
   Four different views regarding a data warehouse design.: the top down view, the data source view, the data warehouse view, and the business query view.
   Top-down view
   - allows selection of the relevant information necessary for the data warehouse

   Data source view
   - exposes the information being captured, stored, and managed by operational systems

   Data warehouse view
   - consists of fact tables and dimension tables

   Business query view
   - sees the perspectives of data in the warehouse from the view of end-user

8. What are the four operations of OLAP?
   Pivoting: changing the dimensions used in a cross-tab
   E.g., moving colors to column names
   Slicing: (sometimes called- dicing) creating a cross-tab for fixed values only
   E.g., fixing color to white and size to small
   Rollup: moving from finer-granularity data to a coarser granularity
   E.g., moving from aggregates by day to aggregates by month or year
   Drill down: The opposite operation of rollup - that of moving from coarser-granularity data to finer-granularity data

9. Why preprocess the data and what are the preprocessing techniques?
   Data preprocessing techniques can improve data quality, thereby helping to improve the accuracy and efficiency of the subsequent mining process. Data preprocessing is an important step in the knowledge discovery process, because quality decisions must be based

on quality data. Detecting data anomalies, rectifying them early, and reducing the data to be analyzed can lead to huge payoffs for decision making. Data preprocessing techniques are
-Data Cleaning
-Data Integration
-Data Reduction
-Data Transformation

10. Data warehouse systems use back-end tools and utilities to populate and refresh their data. Which functions are included in these tools and utilities?

Data extraction
- get data from multiple, heterogeneous, and external sources

Data cleaning
- detect errors in the data and rectify them when possible

Data transformation
- convert data from legacy or host format to warehouse format

Load
- sort, summarize, consolidate, compute views, check integrity, and build indicies and partitions

Refresh
- propagate the updates from the data sources to the warehouse

11. How many categories are there in data cube measure and define each measure with example functions.

Measures can be organized into three categories—distributive, algebraic, and holi stic—based on the kind of aggregate functions used.
Distributive : A measure is distributive if it is obtained by applying a distributive aggregate function. Distributive measures can be computed efficiently because of the way the computation can be partitioned. sum(), count(), min(), and max().
Algebraic: An aggregate function is algebraic if it can be computed by an algebraic function with M arguments (where M is a bounded positive integer), each of which is obtained by applying a distributive aggregate function. avg(), min N() and max N()
Holistic: An aggregate function is holistic if there is no constant bound on the storage size needed to describe a subaggregate. median(), mode(), and rank().

12. What are the most popular data model for a data warehouse and which form exist?

The most popular data model for a data warehouse is a multidimensional model, which can exist in the form of a star schema, a snow flake schema, or a fact constellation schema.

Star schema: A fact table in the middle connected to a set of dimension tables.

Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake.

Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation.